INCH-POUND
MIL-DTL-21604/5B
27 June 2003
SUPERSEDING
MIL-S-21604/5A(SH)
30 March 1992

DETAIL SPECIFICATION SHEET

SWITCH, ROTARY, MULTIPOLE AND SELECTOR, 10 AMPERE,

STYLE JR

This specification is approved for use by the Navy Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-21604.

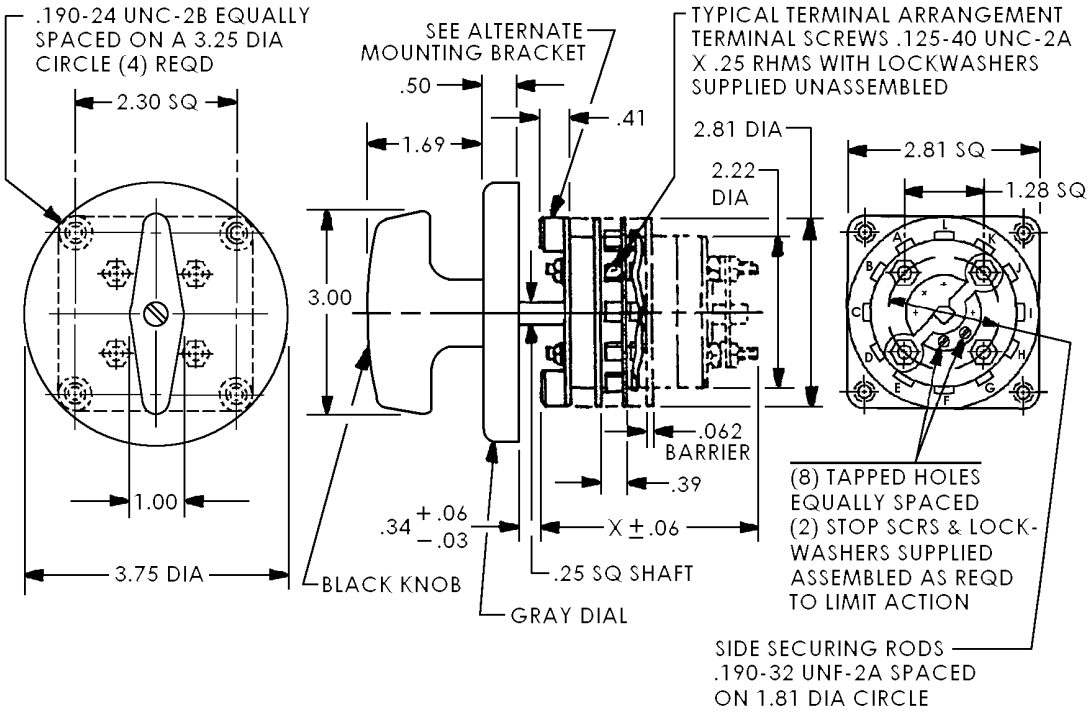
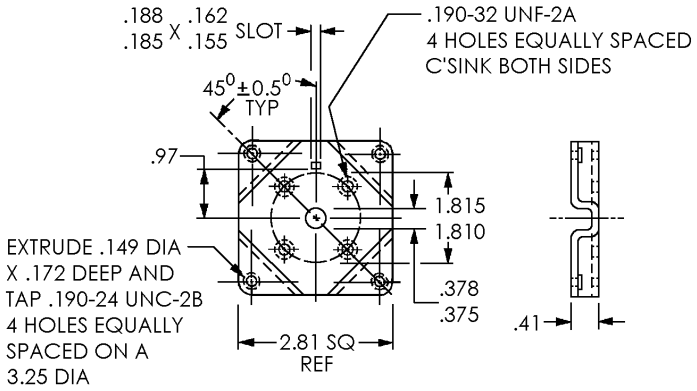


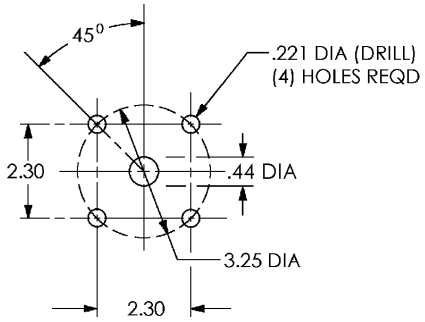
FIGURE 1. Style JR switch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

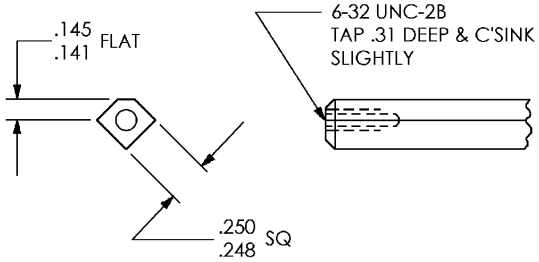
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.



ALTERNATE MOUNTING BRACKET



PANEL MOUNTING DIMENSIONS
AND REAR SUPPORT



SHAFT

FIGURE 1. Style JR switch - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

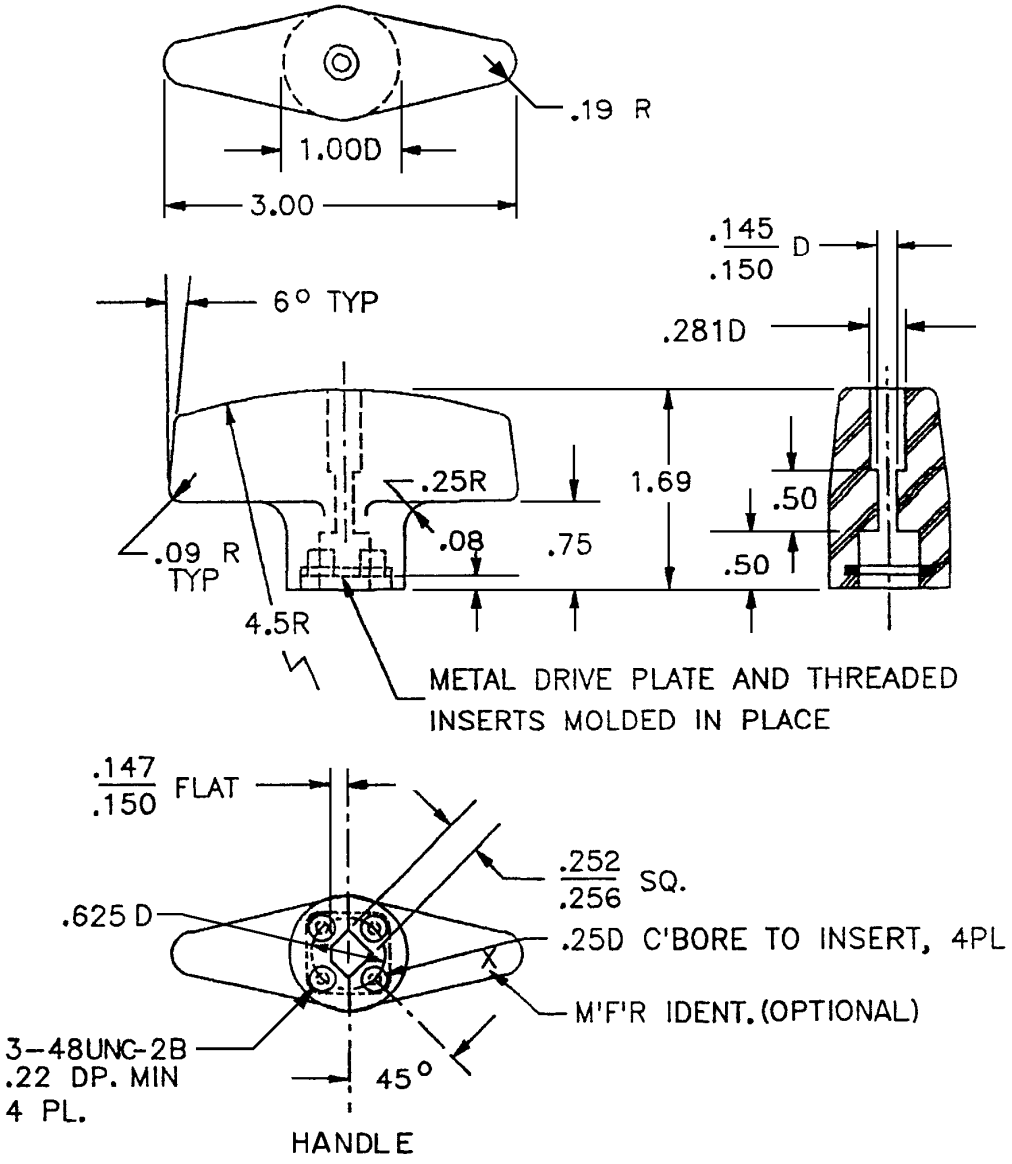


FIGURE 1. Style JR switch - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

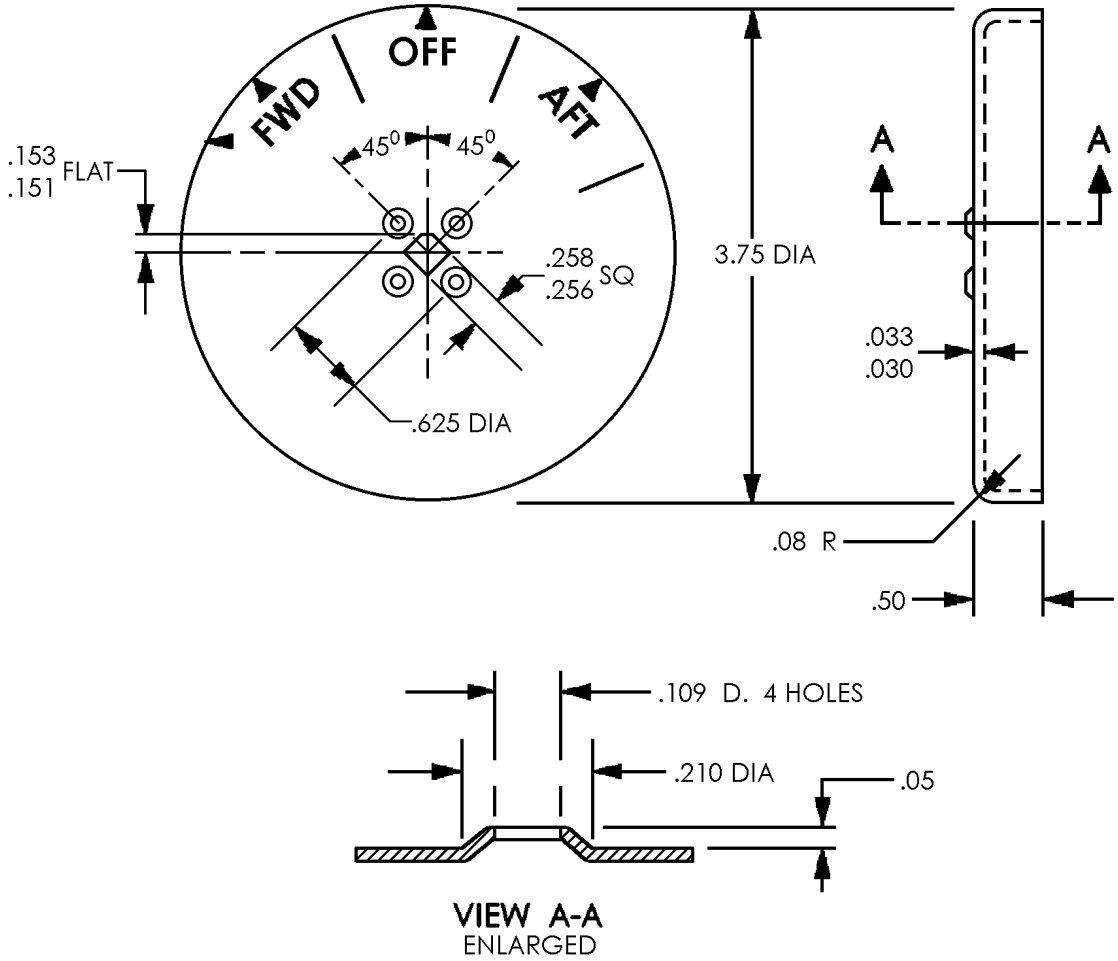


FIGURE 1. Style JR switch - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

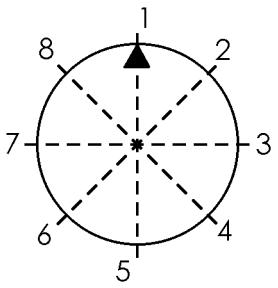
Inch	mm	Inch	mm	Inch	mm
.005	0.13	.185	4.70	.44	11.18
.016	0.41	.188	4.78	.50	12.70
.030	0.76	.19	4.8	.625	15.88
.03	0.8	.190	4.83	.75	19.05
.033	0.84	.210	5.33	.97	24.64
.06	1.52	.22	5.59	1.00	25.40
.062	1.57	.221	5.61	1.28	32.51
.08	2.0	.248	6.30	1.69	42.93
.09	2.3	.25	6.3	1.810	45.97
.109	2.77	.250	6.35	1.81	46.0
.125	3.18	.252	6.40	1.815	46.10
.141	3.58	.256	6.50	2.22	56.39
.145	3.68	.258	6.55	2.30	58.42
.147	3.73	.281	7.14	2.81	71.37
.149	3.78	.31	7.9	3.00	76.20
.150	3.81	.34	8.64	3.25	82.55
.151	3.83	.375	9.52	3.75	95.25
.153	3.89	.378	9.60	4.5	114.3
.155	3.94	.41	10.41	6.0	152.4
.162	4.11				

NOTES:

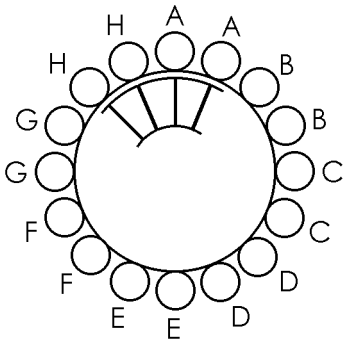
1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .016 (0.41 mm) for two-place decimals and ± 0.005 (0.13 mm) for three-place decimals.
4. On switches of 20 sections or more, the securing rods shall extend .50 inch (12.70 mm) beyond the hex nuts at the rear of the switches to facilitate attachment to rear supports. A hex nut and lockwasher shall be provided for each rod extension.
5. Dial marking, when required, shall be specified in MIL-DTL-21604 (see 6.2).

FIGURE 1. Style JR switch - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

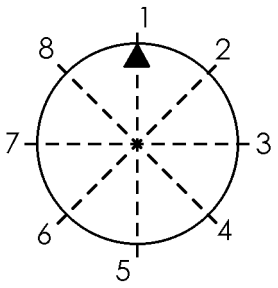


HANDLE POSITIONS

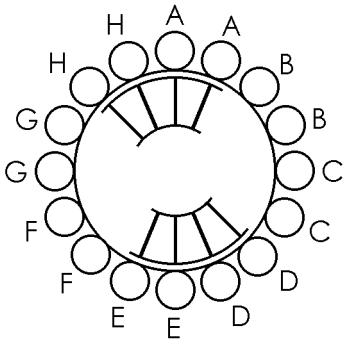


TYPICAL SECTION
SINGLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 1



HANDLE POSITIONS



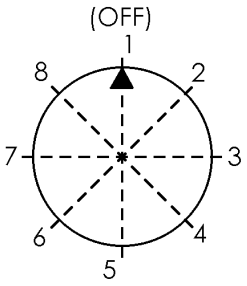
TYPICAL SECTION
DOUBLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 2

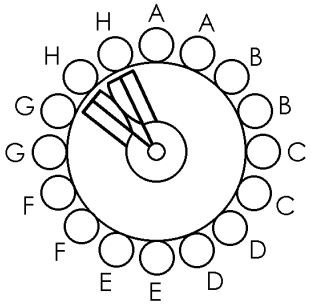
NOTE: Viewed from the handle end.

FIGURE 2. Rotor contact configuration and handle positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

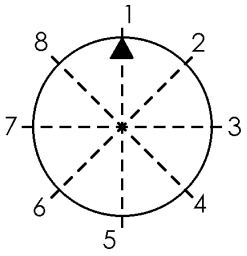


HANDLE POSITIONS

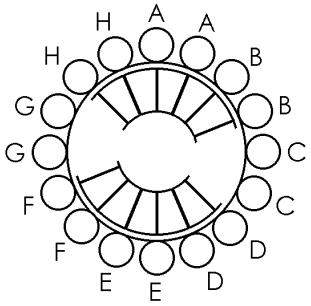


TYPICAL SECTION
SINGLE POLE
SELECTOR

ROTOR DESIGNATION



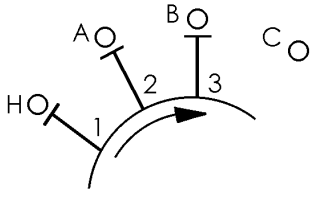
HANDLE POSITIONS



TYPICAL SECTION
DOUBLE POLE
TRIPLE BREAK

ROTOR DESIGNATION 4 1/

1/ WITH ROTOR CONFIGURATION 4, CONTACT H-1 BREAKS BEFORE C-3 MAKES, AND CONTACTS A-1 AND B-2 MAKE BEFORE A-2 AND B-3 BREAK



NOTE: Viewed from handle end.

FIGURE 2. Rotor contact configuration and handle positions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1, figure 2, and table I.
Angle of throw: 45 degrees.
Switching action: Detent.
Electrical loads:
AC current: 10 amperes.
Voltage: 125 volts (at 60 and 400 Hz, 0.80 inductive power factor).
Temperature rise: 30°C maximum.
Contact material:
Rotor: Silver-overlay phosphor bronze.
Stator: Silver-overlay copper.
Barriers: See figure 1.
Terminal marking: See figure 1 and figure 2.
Stop marking: See figure 1.
Qualification samples: See table II.

TABLE I. Type designation and switching characteristics.

Type designation		Style	Number of sections	Dimension "X" 2/ ± .062 (1.57)	Approximate weight (ounces)	Maximum rotational torque (inch-pounds)
Component	Rotor designation 1/					
S	1	JR	5	3.75 (95.25)	29	25
S	1	JR	10	5.62 (142.88)	39	30
S	1	JR	15	8. (203.20)	48	45
S	1	JR	20	9.94 (252.53)	60	50
S	1	JR	25	11.81 (300.02)	70	55
S	2	JR	5	3.75 (95.25)	29	25
S	2	JR	10	5.62 (142.88)	39	30
S	2	JR	15	8. (203.20)	48	45
S	2	JR	20	9.94 (252.53)	60	50
S	2	JR	25	11.81 (300.02)	70	55
S	3	JR	5	3.75 (95.25)	29	25
S	3	JR	10	5.62 (142.88)	39	30
S	3	JR	15	8. (203.20)	48	45
S	3	JR	20	9.94 (252.53)	60	50
S	3	JR	25	11.81 (300.02)	70	55
S	4	JR	5	3.75 (95.25)	29	25
S	4	JR	10	5.62 (142.88)	39	30
S	4	JR	15	8. (203.20)	48	45
S	4	JR	20	9.94 (252.53)	60	50
S	4	JR	25	11.81 (300.02)	70	55

1/ For rotor designation, see figure 2.
2/ Dimensions are in inches. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.

TABLE II. Qualification test sample.

Sample switch	Test number 1/	Sample switch	Test number
S1JR25	1	S3JR25	4
S1JR25	2	S4JR25	1
S1JR25	3	S4JR25	2
S2JR25	4	S4JR25	3
S2JR25	1	S4JR25	4
S2JR25	2	S4JR25	5
S3JR25	3	S4JR25	6

1/ Test numbers in accordance with table II of MIL-DTL-21604.

Custodians:
Navy - SH
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-1749-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:02Z
Check the source to verify that this is the current version before use.